

# A Tabu Search Algorithm for Routing and Spectrum Allocation in Elastic Optical Networks

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## ABSTRACT

In this paper we present a tabu search-based algorithm, which is dedicated to solving the routing and spectrum allocation problem in elastic optical networks. To evaluate performance of the proposed method, we carry a number of experiments on realistic network topologies. We compare our method with both optimal results yielded by the CPLEX solver and results obtained by heuristic algorithms that were proposed in the literature for the same optimization problem. The findings of our investigation show that the proposed tabu search method provides the solutions close to optimal ones and it significantly outperforms the reference methods.

**Keywords:** elastic optical networks, routing and spectrum allocation, tabu search algorithm, optimization.

## 1. INTRODUCTION

Recently, elastic optical networks (EONs) have been proposed to be a successor of the currently deployed wavelength division multiplexing (WDM) networks in future optical transport networks. Thanks to use of flexible frequency grids [1] and advanced modulation and transmission techniques [2], EONs improve the spectral efficiency, scalability, and granularity of optical transport networks when compared to conventional WDM networks.

In EONs a new optimization problem arises which concerns routing and spectrum allocation (RSA). The RSA problem consists in finding a lightpath for each traffic demand, where the lightpath is specified by a routing path (*i.e.*, a sequence of fiber links that connect the source and destination nodes of the demand) and an unoccupied segment of spectrum resources on this path (called a channel). RSA is a basic optimization problem related to the EON design and operation and it was proved to be *NP*-hard [3]. Therefore, many different methods have been proposed to solve it, including ILP formulations [3], [4] and heuristic algorithms [2], [3], [4]. Since metaheuristic methods can be adopted to different optimization problems and are considered as a very efficient method of solving complex problems, in this paper, we propose a novel tabu search-based approach to solve the RSA problem. We run a number of experiments on realistic network topologies and compare our algorithm with reference methods proposed in the literature. The results of our study show that our method significantly outperforms the other algorithms.

The rest of paper is organized as follows. Section 2 includes the ILP model of the RSA problem considered in this paper. The proposed tabu search algorithm is presented in Section 3. Next, Section 4 presents the results of numerical experiments. Finally, the last Section concludes our work.

## 2. PROBLEM FORMULATION

In this Section, we present a format model of the RSA problem studied in this paper.

The elastic optical network is modelled as a directed graph  $G = (V, E)$ , where  $V$  is a set of network nodes and  $E$  denotes a set of fiber links connecting two nodes in  $V$ . On each fiber link, a set of frequency slices is given. Transmission channels are created by grouping an even number of adjacent slices. A set of unicast traffic demands is given. An demand is characterized by a source node, destination node, and traffic volume  $h_d$  (in Gbps). We use the link-path modeling approach [5] and assume  $k$  different routing paths between each pair of nodes. Thus, there are  $|P_d| = k$  candidate paths for each demand. For each routing path, based on its length, we choose a modulation format by applying a rule according to which the number of required regenerators on the path is minimized. Therefore, the selection of modulation formats is not included in the ILP model.

| indices                   |                                                        | constants                                                                                                 | variables                                                                                                  |
|---------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| $e = 1, 2, \dots, E$      | network links                                          | $h_d$ volume (requested bitrate) of demand $d$ (Gbps)                                                     | $x_{dpc} = 1$ , if channel $c$ on candidate path $p$ is used to realize demand $d$ ; 0, otherwise (binary) |
| $s = 1, 2, \dots, S$      | frequency slices                                       | $n_{dp}$ requested number of slices for demand $d$ on path $p$                                            | $y_{es} = 1$ , if slice $s$ is occupied on link $e$ ; 0, otherwise (binary)                                |
| $d = 1, \dots, D$         | unicast demands                                        | $\gamma_{dpcs} = 1$ , if channel $c$ associated with demand $d$ on path $p$ uses slice $s$ ; 0, otherwise | $y_s = 1$ , if slice $s$ is occupied on any network link; 0, otherwise (binary)                            |
| $p = 1, 2, \dots, P_d$    | candidate paths for flows realizing demand $d$         | $\delta_{edp} = 1$ , if link $e$ belongs to path $p$ realizing demand $d$ ; 0, otherwise                  |                                                                                                            |
| $c = 1, 2, \dots, C_{dp}$ | candidate channel for demand $d$ allocated on path $p$ |                                                                                                           |                                                                                                            |

**objective**

$$\min \varphi = \sum_s y_s \quad (1)$$

**subject to**

$$\sum_p \sum_c x_{dpc} = 1, d = 1, 2, \dots, D \quad (2)$$

$$\sum_d \sum_p \sum_c \gamma_{dpcs} \delta_{edp} x_{dpc} = y_{es}, e = 1, 2, \dots, E, s = 1, 2, \dots, S \quad (3)$$

$$y_{es} \leq y_s, e = 1, 2, \dots, E, s = 1, 2, \dots, S \quad (4)$$

The ILP model is described by formulas (1)-(4). The optimization objective, defined as a maximum spectrum usage, is described by equation (1). Constraint (2) assures for each demand  $d$  exactly one candidate lightpath is selected. Finally, equations (3) and (4) control spectrum usage.

### 3. TABU SEARCH ALGORITHM

The proposed tabu search algorithm (called in the paper as “TS”) is based on the standard procedure initially proposed by Glover [6]. Since the solution space in the RSA problem is huge and the evaluation of a particular solution is time-consuming (calculating value of the objective function), we propose a method that does not implement a local search mechanism in each iteration, similarly like in our previous work [7]. The base idea of the proposed TS method is presented in Fig. 1.

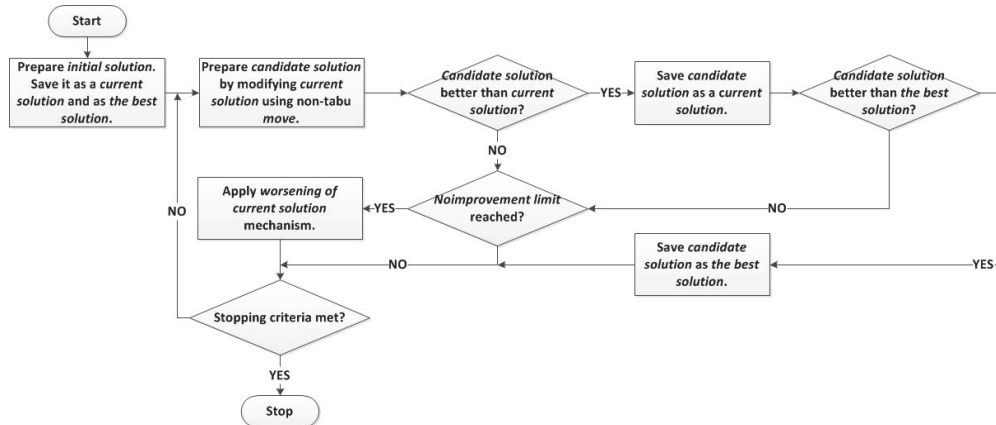


Figure 1. The tabu search algorithm.

In our method, a solution corresponds to a set of allocation rules that include the demand allocation order (demands are allocated one by one according to this order) and the path selection (which indicates a candidate path selected for each traffic demand). The demands are allocated with respect to these solution rules and a first-fit spectrum assignment method. Concurrently, a move operation is a definition of a single change in a solution. In particular, the move operation can be defined as a demand order swap or a path swap. The first operation chooses two demands and swap their positions in the demand allocation order, wherein the second operation changes the selected path for one traffic demand.

At the beginning, the TS method finds an initial solution by means of the modified heuristic algorithm called Adaptive Frequency Assignment (AFA) that was proposed in [3]. The obtained solution is stored as a current solution and as the best found solution. Then, the algorithm iteratively searches through the feasible solution space, in order to find a better solution. In this end, in each iteration a non-tabu (non-forbidden) move operation is generated and applied to modify the current solution. The result of this modification is a candidate solution. If the candidate solution is better than the current, it is stored as a new current solution. Moreover, if this solution is also better than the already best found solution, it is stored as the new best found solution as well. If the candidate solution is worse than the best solution, no improvement was reached in current iteration.

The TS search process is supported by a diversification process that is applied when the number of iterations with the no improvement status exceeds a predefined threshold. This threshold is defined as a ratio of the number of all traffic demands and is an input parameter to the algorithm. For example, if the no improvement threshold is equal to 20% and  $D = 100$ , then the diversification process is run after 20 iterations without improvement. The diversification mechanism multiplies the value of the objective function related to the current solution by a coefficient  $\beta$  ( $\beta > 1$ ). The value of the  $\beta$  parameter is an input parameter to the algorithm. By these means, the TS method can accept even a weaker solution as the new current solution in next iterations, and thus change the evaluated part of the solution space.

The move generation process is controlled by two lists of forbidden moves: a tabu list (TL) and a used moves

(UM) list. The forbidden moves are operations that cannot be applied in the next iteration, since they can lead to solutions already considered. The TL is implemented as a FIFO queue. Its size is a ratio of the number of all traffic demands and is an input parameter to the algorithm. For example, if the TL size is equal to 20% and  $D = 100$ , then TL can store at most 20 move operations. The TL is a set of moves, which led to some improvement in previous iterations. Concurrently, the UM list is a set of moves, which did not lead to any improvement since the last improvement. The size of this list is not limited. The UM list is cleared after an improvement or during a diversification process.

#### 4. NUMERICAL EXPERIMENTS

In our study, we focus on three different network topologies: NSF15 (15 nodes, 46 links) [8], Euro16 (16 nodes, 48 links) [8] and US26 (26 nodes, 84 links) [9]. The source and destination nodes of traffic demands are generated at random and the demand volume is uniformly distributed between 10 and 200 Gbps.

We assume that the considered EON operates with bandwidth-variable transponders (BV-Ts) implementing the PDM-OFDM technology and supporting six modulation formats selected adaptively, *i.e.*, BPSK, QPSK, 8-QAM, 16-QAM, 32-QAM, 64-QAM, with the spectral efficiency equal to 1, 2, ..., 6 b/s/Hz, respectively. The transmission range of the selected modulation format is a function of the modulation level and the demand bit-rate, similarly as in the model presented in [10]. The EON implements a flexible ITU-T grid of a 6.25 GHz granularity [1]. We assume a mixed-line-rate approach with three different types of BV-Ts, supporting bit-rates of up to 40 Gbps, 100 Gbps, and 400 Gbps. We introduce a 12.5 GHz guard band between neighboring lightpaths. In all scenarios, we assume that the transmission reach can be extended by means of regenerators, if the path length exceeds the transmission range of the selected modulation format.

##### 4.1 Tuning of the TS algorithm

In order to find the settings of the TS algorithm that assure good algorithm performance, a tuning process was performed. To this end, for each network topology, we evaluated a group of 5 different testing scenarios and about 100 combinations of input parameters. Next, for each setting, we run 10 times the TS algorithm and, eventually, we compared the averaged results selecting the best setting for further experiments.

In Table 1, we present the range of each parameter considered during the algorithm tuning as well as the values corresponding to the best setting. Due to space limitations, we do not present detailed tuning results.

Table 1. Values of the TS input parameters analyzed in the tuning process and selected for further experiments.

| Parameter            | Range of evaluated values                 | Best settings |        |      |
|----------------------|-------------------------------------------|---------------|--------|------|
|                      |                                           | NSF15         | Euro16 | US26 |
| No improvement limit | 0% – 200%, with 10% step                  | 160%          | 180%   | 200% |
| Tabu List size       | 0% – 200%, with 10% step                  | 50%           | 160%   | 10%  |
| $\beta$              | 1.10 – 1.40, with 0.02 step               | 1.24          | 1.16   | 1.36 |
| Number of iterations | 25000, result stored every 100 iterations | 12000         | 8000   | 1000 |

##### 4.2 Comparison with other methods

The evaluation of the efficiency of the proposed TS method was performed based on the comparison with four different heuristic algorithms, proposed in the literature to solve the same optimization problem. In particular, we implemented the following reference algorithms: First First (FF) [2], Most Subcarriers First (MSF) [4], Longest Path First (LPF) [4], and Adaptive Frequency Assignment (AFA) [3].

First, we compared the algorithms performance with optimal results yielded by the CPLEX solver (ver. 12.5). The evaluation was performed for NSF15 and Euro16. We used a group of 50 different testing scenarios. Each testing scenario is an unique set of unicast demands, where the total traffic volume in the network is equal to 3 Tbps. Moreover, for all scenarios we assumed the number of candidate paths  $k = 2$ . Since TS is not a deterministic method, for each testing scenario, we run 30 times the algorithm and averaged the results. For each evaluated method, the average optimality gap (vs. CPLEX) is presented in Table 2. We can see that the TS method significantly outperforms the reference heuristics. Concurrently, in Fig. 2 we present the percentage of optimal solutions obtained with each method. The proposed TS methods founds optimal

Table 2. Comparison of algorithms: average gap to optimal solution.

|        | FF     | MSF    | LPF    | AFA   | TS    |
|--------|--------|--------|--------|-------|-------|
| NSF15  | 35.56% | 17.77% | 17.55% | 7.68% | 1.13% |
| Euro16 | 34.26% | 17.77% | 15.72% | 7.37% | 1.38% |

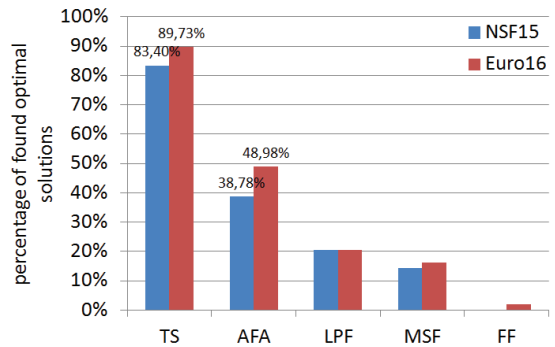


Figure 2. Percentage of optimal solutions obtained with each tested algorithm.

solutions (*i.e.*, such as provided by CPLEX) in more than 80% of all simulations.

In the above experiments, the processing time of heuristics and the TS method was less than 1 ms and 10 s, respectively. The CPLEX solver needed on average 58 s to solve the same problem instances.

Next, we used TS and other heuristics for solving larger problem instance. Here, the CPLEX solver was not able to reach the optimality within a time limit of 1 hour. We used a group of 30 different testing scenarios with the total traffic volume equal to: 30 Tbps (NSF15, Euro16) and 50 Tbps (US26). Moreover, we performed the evaluation for different numbers of candidate paths (between each pair of nodes), *i.e.*,  $k = 2, 5, 10, 30$ . For each testing scenario, the TS algorithm was run 5 times and the results were averaged. In Table 3, we report an average gap to the best found solution (among all tested methods). Once again, the proposed TS method yielded the best results.

In this part of investigation, the reference heuristics were able to solve the problem in less than 2 s. The TS method needed on average between 14 s (NSF15, Euro16) and 114 s (US26). Comparing to the CPLEX solver, which was not able to solve the problem in an 1-hour period, these values are relatively small.

Table 3. Comparison of algorithms: average gap to best found solution.

| Scenario |          | FF     | MSF    | LPF    | AFA   | TS    |
|----------|----------|--------|--------|--------|-------|-------|
| NSF15    | $k = 2$  | 36.76% | 20.93% | 16.60% | 7.66% | 2.38% |
|          | $k = 5$  | 46.21% | 21.05% | 19.11% | 4.41% | 2.05% |
|          | $k = 10$ | 47.08% | 19.31% | 20.98% | 3.05% | 1.45% |
|          | $k = 30$ | 46.86% | 20.45% | 24.14% | 1.77% | 1.02% |
| Euro16   | $k = 2$  | 40.45% | 16.20% | 15.07% | 6.11% | 1.66% |
|          | $k = 5$  | 52.40% | 22.08% | 21.93% | 3.10% | 1.29% |
|          | $k = 10$ | 52.82% | 19.59% | 19.75% | 2.80% | 1.61% |
|          | $k = 30$ | 52.06% | 19.90% | 22.88% | 2.41% | 1.12% |
| US26     | $k = 2$  | 35.22% | 17.80% | 17.29% | 3.53% | 0.36% |
|          | $k = 5$  | 53.90% | 19.71% | 16.79% | 2.72% | 1.39% |
|          | $k = 10$ | 58.06% | 19.33% | 18.01% | 1.38% | 0.96% |
|          | $k = 30$ | 59.46% | 19.63% | 21.61% | 1.02% | 0.51% |

### 4.3 Efficiency of the move operations

Finally, in Table 4 we present an extension of the NSF15 results presented in Table 3. In particular, we focus on three different versions of the TS method: 1) the TS that implements only the path swap (PS) move, 2) the TS that implements only the demand swap (DS) move, and 3) a standard TS with both types of moves allowed. The obtained results show that the use of both types of operations in TS yields the best algorithm performance.

Table 4. Average TS gap to best found result for NSF15 and different allowed move operations.

|          | PS move | DS move | PS and DS move |
|----------|---------|---------|----------------|
| $k = 5$  | 2.12%   | 2.79%   | 2.05%          |
| $k = 10$ | 1.61%   | 1.79%   | 1.45%          |
| $k = 30$ | 1.34%   | 1.08%   | 1.02%          |

## 5. CONCLUSIONS

In this paper, we have proposed a novel method dedicated to solve the routing and spectrum allocation problem in elastic optical networks. The algorithm is based on the tabu search approach. We performed an extensive set of numerical experiments on three realistic network topologies to examine the efficiency of our method and compared it with reference heuristic methods proposed in the literature. The findings show that the proposed method obtains very good results and significantly outperforms the reference algorithms.

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## REFERENCES

- [1] ITU-T Recommendation G.694.1 (ed. 2.0), "Spectral grids for WDM applications: DWDM frequency grid," Feb. 2012.
- [2] M. Jinno *et al.*, "Distance-adaptive spectrum resource allocation in spectrum sliced elastic optical path network," *IEEE Communications Magazine*, vol. 48, no. 8, 2010, pp. 138-145.
- [3] M. Klinkowski and K. Walkowiak, "Routing and spectrum assignment in spectrum sliced elastic optical path network," *IEEE Communications Letters*, vol. 15, no. 8, 2011.
- [4] K. Christodoulopoulos *et al.*, "Elastic bandwidth allocation in flexible OFDM-based optical networks," *IEEE Journal of Lightwave Technology*, vol. 29, no. 9, 2011, pp. 1354-1366.
- [5] M. Pióro and D. Medhi, "Routing, Flow and Capacity Design in Communication and Computer Networks," Morgan Kaufmann, 2004.
- [6] F. Glover, "Tabu Search Fundamentals and Uses," University of Colorado, June 1995.
- [7] K. Walkowiak *et al.*, "On minimization of the spectrum usage in elastic optical networks with joint unicast and anycast traffic," in *Proc. ACP*, 2013.
- [8] K. Walkowiak and M. Klinkowski, "Joint anycast and unicast routing for elastic optical networks: Modeling and optimization," in *Proc. IEEE ICC*, 2013.
- [9] R. Gościński *et al.*, "Joint anycast and unicast routing and spectrum allocation in elastic optical networks with dedicated path protection consideration," in *Proc. DRCN*, 2014.
- [10] C. Politi *et al.*, "Dynamic operation of flexi-grid OFDM-based networks," in *Proc. OFC*, 2012.